LISFLOOD-FP 8.4: portable multi-GPU flood solvers with hydrological losses and optimised pluvial-flood CUDA solvers, validated on real cases and against three other GPU solvers

Haseeb ur Rehman[1], Félicia Norma Teferle[1], Georges Kesserwani[2]

[1]University of Luxembourg, Luxembourg
[2]University of Sheffield, United Kingdom

Key points

- LISFLOOD-FP 8.4 adds infiltration and evaporation to every graphics-processor solver, on NVIDIA and Apple hardware
- Uncalibrated Green-Ampt soil parameters raise the mean Nash-Sutcliffe efficiency at four Alzette gauges from 0.12 to 0.66
- The optimised solvers run 1.3 to 3.3 times faster with identical results and are the fastest of four graphics-processor flood solvers

Corresponding author: Haseeb ur Rehman, `haseeb.rehman@uni.lu`

Abstract

LISFLOOD-FP 8.4 adds constant-rate, Green–Ampt and Horton infiltration and evaporation to all seven CUDA solvers and to native Apple Metal uniform-grid FV1, FV2, DG2 and ACC solvers. The uniform-grid FV1, FV2 and DG2 solvers run on several GPUs, and the CUDA FV1 and ACC steps are made 1.3–3.3 times faster than in 8.3 with identical predictions. The losses reproduce analytical solutions to within 1 mm, FV1 gives identical results on one and four GPUs, and on four real FloodCastBench events the stored water ends within 4.4 % of the references. For the July 2021 pluvial flood of the upper Alzette (4.7 million cells), CUDA FV1 and three other first-order finite-volume CUDA solvers, HiPIMS, SERGHEI-SWE and TRITON, ran on the same inputs and hardware; all overestimate the peaks without infiltration. Uncalibrated Green–Ampt parameters from European soil databases raise the mean Nash–Sutcliffe efficiency at four gauges from 0.12 to 0.66 (HiPIMS: 0.71); Horton gives 0.64; the time-based Horton form of SERGHEI-SWE infiltrates half as much, its capacity decaying before the storm. CUDA FV1 is the fastest on one A100 (33 min); its kernels are memory-bound and four GPUs give a speed-up of 1.2–1.9.

Plain Language Summary

Flood simulations on graphics processors can now cover whole river basins at a resolution of a few metres, but most of them treat the ground as waterproof: every drop of rain stays on the surface. In real storms part of the rain soaks into the soil, so these simulations overestimate the flood. Version 8.4 of the open-source flood model LISFLOOD-FP lets the soil absorb water and the air evaporate it in all its graphics-processor versions, on NVIDIA cards and on Apple laptops, and lets one simulation run on several graphics processors at once. We check the new code against exact solutions, against four large published flood simulations, and against measured river flows during the July 2021 flood of the Alzette river in Luxembourg, where we also run three other graphics-processor flood models on the same inputs. All four overestimate the flood peaks when the soil is waterproof. Soil properties taken from European databases, without any tuning, bring the simulated river flows much closer to the measurements. The improved code also runs up to three times faster than the previous version, gives exactly the same results, and is the fastest of the four models tested.

1 Introduction

LISFLOOD-FP solves the two-dimensional shallow-water equations with a local-inertial scheme (ACC; Bates, Horritt, & Fewtrell, 2010), a first-order finite-volume scheme (FV1), a second-order finite-volume scheme (FV2) and a second-order discontinuous Galerkin scheme (DG2), with GPU versions of these solvers and of three variants on non-uniform and adaptive meshes (Shaw et al., 2021; Sharifian et al., 2023; Chowdhury & Kesserwani, 2025; LISFLOOD-FP developers, 2024). Version 8.3 made the uniform-grid CUDA solvers faster and reproducible and added an Apple Metal back end for FV1, so that the July 2021 flood of the upper Alzette (Sect. 4.2) could be simulated competitively on an Apple laptop instead of an NVIDIA A100 of a computer cluster. The developments of this paper stem from the same application, the simulation of pluvial floods at catchment scale from rainfall on a 10 m grid, and the CUDA FV1 solver that serves it is the solver carried through the validation of Sect. 4.2.

Three limitations remained, and this paper addresses them in turn. First, infiltration and evaporation were applied only by CPU solvers: a uniform infiltration rate by the local-inertial solver and a per-cell rate by the sub-grid channel model, which 8.4 leaves unchanged. The FV1 and DG2 solvers on the CPU and every GPU solver read the loss keywords and ignored them, so a GPU flood simulation was purely hydraulic. This can be a large omission in pluvial floods, such as the catchment-scale event simulated in Sect. 4.2, where infiltration removes about one third of the rain. Second, on Apple computers FV2, DG2 and ACC ran only on the CPU. Third, every GPU simulation ran on a single device, which limits the grid size and the turnaround of large domains.

Version 8.4 adds three infiltration models (a constant rate, Green–Ampt and Horton) and evaporation to all seven CUDA solvers, ports FV2, DG2 and ACC to Metal with the same losses, distributes FV1, FV2 and DG2 over several GPUs (Table 1) and makes the CUDA FV1 and ACC solvers faster. These two are the solvers used for the real events of Sect. 4: FV1 solves the full shallow-water equations, which rain-driven flow over steep terrain requires, and ACC is the local-inertial solver of operational floodplain modelling. While testing the losses we also found and corrected a free outflow boundary that admitted water when the flow at the

edge pointed into the domain. Section 2 describes these developments and their use. Section 3 verifies them against analytical solutions, against the previous release and across back ends and GPU counts. Section 4 evaluates the accelerated CUDA ACC and FV1 solvers on large real pluvial floods: first the hydraulics, against the reference simulations of the FloodCastBench dataset (Xu et al., 2025a), and then the infiltration models, with CUDA FV1 alone, for the July 2021 flood of the upper Alzette, against observed discharge and against three independent GPU solvers run on the same inputs and hardware: HiPIMS (Xia et al., 2019), SERGHEI-SWE (Caviedes-Voullième et al., 2023) and TRITON (Morales-Hernández et al., 2021). All three are, like FV1, first-order finite-volume solvers written in CUDA, so the comparison is like for like and isolates the loss models from the hydraulic scheme. Section 5 reports the computational cost and the hardware limits of the solver, and Sect. 6 states what remains unfinished.

2 Developments in version 8.4

2.1 Infiltration and evaporation on the GPU

The losses are evaluated cell by cell on the GPU once per time step: after the hydraulic update in FV1, FV2 and DG2, and, in ACC, at the point where the CPU ACC solver applies them (after the dry check and before the depth update). Cells without data are skipped. One infiltration model is chosen in the parameter file; evaporation, a time series in $\mathrm{mm\,day^{-1}}$ that is uniform in space as in the CPU solvers, can be combined with any of them. Every parameter of the three infiltration models is either a number or a grid file.

Constant rate As in the CPU solvers, every cell whose water depth exceeds the drying threshold (keyword `depththresh`, 1 mm by default, below which a cell is treated as dry) loses $(i + e)\,\Delta t$, where i is the infiltration rate (keyword `infiltration` in $\mathrm{m\,s^{-1}}$, or a grid `infilfile` in $\mathrm{mm\,h^{-1}}$), e the evaporation rate and Δt the time step. The water depth never falls below zero.

Green–Ampt A cell with water infiltrates at

$$f = K_s \left(1 + \frac{(\psi + h_0)\,\Delta\theta}{F}\right), \tag{1}$$

92 **Table 1.** GPU capabilities added in 8.4. "Losses" denotes constant-rate (uniform or gridded),
93 Green–Ampt or Horton infiltration, each with optional evaporation. Metal was absent in 8.2; in
94 8.3 it provided FV1 without losses. Dense modes run the Metal FV1, DG2 or ACC operator on
95 every cell of the finest grid and provide no adaptive mesh.

GPU solver	CUDA in 8.4	Metal in 8.4	Several GPUs in 8.4
FV1	losses added	native (from 8.3); losses added	CUDA and Metal
FV2	losses added	new native solver with losses	CUDA and Metal[a]
DG2	losses added	new native solver with losses	CUDA and Metal[a]
ACC	losses added	new native solver with losses	no
ACC_NUGRID	non-uniform mesh; losses added	dense ACC mode with losses	no
HWFV1	adaptive mesh; losses added	dense FV1 mode with losses	no
MWDG2	adaptive mesh; losses added	dense DG2 mode with losses	no

[a] Scaling not yet measured; on Metal all ranks share one GPU (Sect. 6).

111 where K_s is the saturated hydraulic conductivity, ψ the soil suction head at the
112 wetting front, $\Delta\theta$ the soil capacity (porosity minus initial moisture content), h_0
113 the depth of ponded water and F the cumulative infiltration (Green & Ampt,
114 1911; Mein & Larson, 1973); these are the parameters of the model as used in
115 flood modelling software such as TUFLOW (BMT, 2025). The increment over
116 a time step, $\Delta F = F' - F$, is calculated from the implicit Euler step $F' = F + k\,(1 + M/F')$, with $k = K_s\Delta t$ and $M = (\psi + h_0)\Delta\theta$, which has the closed
117 form $F' = \frac{1}{2}\big((F + k) + \sqrt{(F+k)^2 + 4kM}\big)$. The increment may not exceed the
118 water in the cell nor, if a soil depth D is given, the storage that remains in the soil,
119 $D\,\Delta\theta - F$.
120

Horton Horton's infiltration capacity decays from an initial rate f_0 to a final rate f_c with a rate constant k_H (Horton, 1941). In its original form the decay follows elapsed time, so the capacity of a cell falls even while the cell is dry. Version 8.4 uses the cumulative form adopted in the Storm Water Management Model (Rossman & Huber, 2016), in which the capacity depends on the depth already infiltrated: the cell is assigned the equivalent time t_e, the time under continuous ponding at which the cumulative infiltration would reach its F, from

$$F = f_c\, t_e + \frac{f_0 - f_c}{k_H}\left(1 - \exp(-k_H t_e)\right), \tag{2}$$

solved by Newton iteration, and may take up to $F(t_e + \Delta t) - F(t_e)$ of the water present in the time step. Under continuous ponding this reproduces the classical Horton curve exactly, and a cell that receives no water keeps its capacity. Green–Ampt and Horton therefore share the same state, the cumulative infiltration F, and the same treatment of drying; in neither does the soil drain and regain capacity between rain events, as the recovery option of SWMM allows.

Single precision The single-precision builds store the water depth in 32-bit floats, and the loss per time step is often close to the resolution of the depth (a few micrometres on metres of water). On both back ends, each cell therefore carries the part of its loss that rounding discarded to its next time step: the depth after the loss is subtracted from the depth before it, a difference that is exact by Sterbenz's lemma, and the shortfall against the intended loss is added to the next step (compensated subtraction, Kahan, 1965). The cumulative infiltration F needs more precision than the depth, because it grows over the whole simulation by increments that stay small. CUDA stores F as a 64-bit double, which NVIDIA GPUs support in every build. Apple GPUs have no double precision, so Metal stores F as a pair of floats updated with an exact two-sum (Dekker, 1971) and computes the Green–Ampt increment without the cancellation in $F' - F$:

$$\Delta F = \begin{cases} \tfrac{1}{2}\left((k - F) + \sqrt{S}\right), & F \le k, \\ \dfrac{2\,(Fk + kM)}{\sqrt{S} + F - k}, & F > k, \end{cases} \qquad S = (F + k)^2 + 4kM. \tag{3}$$

Application in each solver The loss changes the stored variables of each solver in the way that keeps its flow velocity: FV1 scales its discharges with the depth; ACC changes only the depth, because its discharges are stored on the cell

faces; FV2 and DG2 lower the mean depth, keep the depth slopes, so that a flat water surface stays flat, and scale the discharge coefficients, and a cell whose face would become dry is reduced to first order. Whatever the solver, the volume removed is summed on the GPU and reported in the `.mass` file, whose budget now closes on every CUDA and Metal solver on one rank (Sect. 6): it counts the rainfall, sums the stored volume over the cells with data only and reports the cumulative rainfall minus losses in its last column.

Adaptive and non-uniform meshes The CUDA MWDG2 and HWFV1 solvers change their mesh every step, and ACC_NUGRID uses cells of several sizes. Their soil parameters and state (F and the rounding residual) are kept on the finest grid, so they persist when cells merge or split. An active cell evaluates the loss for each of its valid finest children, with that child's parameters and state, at the active cell's mean depth, and removes the mean of the resulting losses. Parameters are thus never averaged before they enter the nonlinear infiltration models, and the `.mass` budget above applies unchanged.

2.2 Free boundary restricted to outflow

A `FREE` boundary is intended to let water leave the domain. In the CPU FV1 and DG2 solvers and in the CUDA and Metal FV1, FV2 and DG2 solvers, the ghost cell of a free boundary copied the state of the adjacent cell, including its velocity. Where the flow at the edge pointed into the domain, the boundary therefore supplied water. Version 8.4 reflects the inward normal component of the discharge in the ghost cell, so that the boundary passes outflow only; the CPU FV1 ghost cell keeps its tangential component. The local-inertial solvers keep their own free boundary, in which the outflow follows a friction law with the local or a prescribed water-surface slope.

2.3 Native Metal solvers

Version 8.4 adds native uniform-grid FV2, DG2 and ACC solvers to Metal and applies the loss models of Sect. 2.1 to them and to FV1 (Table 2). The FV2 and DG2 kernels are ports of the CUDA solvers; the ACC kernels follow the CPU ACC solver. Three design choices determine their correctness. First, FV2 adopts the

Table 2. Native Metal solvers added in 8.4 and their loss implementation.

Solver	Design
FV2	port of CUDA FV2: reconstruction, limiter, bed source term and two Runge–Kutta stages, with the boundary states and step order of CUDA
DG2	port of CUDA DG2: nine coefficients per cell in one buffer of nine planes (Metal allows 31 buffers per kernel), friction at the Gauss points, Krivodonova limiter, slopes of the DEM and the start file
ACC	follows the CPU ACC solver; a conservative two-pass limiter for drying cells keeps each face flux shared by its two cells
Losses	shared kernels for the four native solvers and the dense modes; the same models as CUDA, with the soil state held as a pair of floats (Sect. 2.1)

states of the ghost cells and the step order of CUDA, which a fixed-discharge inflow requires for the boundary to conserve water. Second, as for FV1 in 8.3, each face flux is computed once and shared by its two cells, and the kernels are compiled without the fast-math option, so every run repeats bit for bit. Third, CUDA DG2 limits the slopes in place, so that a cell may read the slope of a neighbour before or after it is limited; Metal writes the limited slopes to a separate buffer and has no such race.

The keywords `ACC_NUGRID`, `HWFV1` and `MWDG2` select dense Metal modes that apply the ACC, FV1 or DG2 operator, with losses, on every cell of the finest grid. A parameter file written for the non-uniform or adaptive CUDA solvers therefore runs on Apple hardware, but at the memory and run time of the uniform solver on the finest grid: the non-uniform and adaptive meshes themselves exist only in CUDA, and porting them was outside the scope of this release.

2.4 Execution on several GPUs

FV1, FV2 and DG2 can run on any number of GPUs of one node, with one rank of the Message Passing Interface (MPI) per GPU; the runs of this paper use up to four, the GPUs of one node of the cluster used. The raster is divided into contiguous bands of rows. Each rank advances its own rows, exchanges the halo rows that the stencil of the solver requires after every stage, and takes part in a global

reduction of the time step. Output is gathered on the first rank in a fixed global order, so that results can be compared directly with a run on one GPU. The same decomposition is used by Metal, where all ranks share the GPU of the computer; there it serves to verify the decomposition rather than to accelerate the run. ACC, the non-uniform `ACC_NUGRID` solver and the adaptive `HWFV1` and `MWDG2` solvers run on one GPU and stop with an error message if more are requested: distributing their meshes over several devices is outside the scope of this release.

2.5 Parameter-file keywords

The new options are keywords in the parameter file; terrain, boundary, rainfall and output settings are unchanged. A CUDA FV1 simulation on four GPUs with Horton infiltration and evaporation reads:

```
cuda
fv1
GPUs 4
infiltration horton
horton_f0 f0.asc
horton_fc fc.asc
horton_k 2
evaporation evap.dat
```

Here f_0 and f_c are grids in $\mathrm{mm\,h^{-1}}$ and k_H a number in $\mathrm{h^{-1}}$. For Green–Ampt, the line `infiltration greenampt` is followed by `greenampt_ks` ($\mathrm{mm\,h^{-1}}$), `greenampt_psi` (mm), `greenampt_porosity`, `greenampt_theta_init` (volumetric fractions) and optionally `greenampt_soil_depth` (m). A constant rate is given in $\mathrm{m\,s^{-1}}$ (`infiltration 5.6e-7`) or as a grid in $\mathrm{mm\,h^{-1}}$ (`infilfile`). On Apple Silicon, `cuda` is replaced by `metal`. A run on N GPUs is started with `mpirun -np N`, where N equals the value of `GPUs`.

3 Verification

All tests are distributed with the code. They ran on an NVIDIA A100 (CUDA 12.6) and an Apple M1 Max (macOS 27). Besides tests written for 8.4, they use the test cases of the 8.2 source (LISFLOOD-FP developers, 2024): three humps

(T01–T03; Liang & Borthwick, 2009; Shirvani, Kesserwani, & Richmond, 2021), a dam break through an idealised city (T05; Soares-Frazão & Zech, 2008), the torrential valley flood of test 5 of the Environment Agency benchmark (T06; Néelz & Pender, 2013; Ayog, Kesserwani, Shaw, Sharifian, & Bau, 2021), the fluvial Carlisle 2005 flood (T07, and T07s for its first 6 h; Horritt, Bates, Fewtrell, Mason, & Wilson, 2010; Neal, Keef, Bates, Beven, & Leedal, 2013), dynamic rain (T030) and an irregular free boundary (T031).

3.1 Loss models against analytical solutions

Three groups of tests isolate the loss models from the hydraulics (Table 3, Fig. 1). T012 is the evaporation and infiltration test of LISFLOOD-FP: a flat basin holding 0.4464 m of water loses 3.6 mm day^{-1} to evaporation and 10^{-8} m s^{-1} to infiltration for 100 days, with variants in which the evaporation rises linearly and in which a wall separates two halves with different rates. The expected depth is the initial depth minus the sum of rate times step over the steps each solver took. In the Green–Ampt test, 0.1 m of water ponds on dry soil ($K_s = 10\,\mathrm{mm\,h^{-1}}$, $\psi = 110\,\mathrm{mm}$, $\Delta\theta = 0.23$) for 24 h; the model is compared with the implicit Euler solution for the steps it took and with the continuous solution. Variants give the parameters as grids, add a soil depth of 0.2 m and set $K_s = 0$ on one half. In the Horton test, 0.2 m of water ponds in a closed box ($f_0 = 60\,\mathrm{mm\,h^{-1}}$, $f_c = 10\,\mathrm{mm\,h^{-1}}$, $k_H = 2\,\mathrm{h^{-1}}$); the water never runs out, so the infiltrated depth follows the classical Horton curve exactly.

All 77 checks on Metal and 76 of the 81 on CUDA pass. The other five compare against assumptions rather than solutions: four find the `.mass` log of 8.4 different from that of 8.3, whose budget it corrects (the depth grids are identical), and one expects 60 s steps. The CUDA FV2 and DG2 errors in T012, about 0.2 % of the loss, arise because these solvers store the bed elevation in single precision as given: at 20 m the last digit is 1.9 μm, about the loss of one step, so per-cell rounding creates small spurious water-surface slopes. With the bed lowered to 0 m they are exact (0.0002 mm), and Metal, which shifts the bed by its lowest elevation, is exact on the original test. Because changing this storage would change CUDA FV2 and DG2 results without the new keywords, 8.4 leaves it as in 8.3. The difference between the model and the continuous Green–Ampt solution (Fig. 1b) is the error of implicit Eu-

Table 3. Loss tests: largest depth error against the expected depth for FV1 / FV2 / DG2 / ACC on CUDA (A100) and Metal (M1 Max), single precision. "Exact" means within the last written digit (0.001 mm).

Test	CUDA (mm)	Metal (mm)
T012, constant rate, 50 days (223 mm lost)	0.0001 / 0.29 / 0.15 / 0.0000	0.0001 / 0.0002 / 0.0002 / 0.0000
T012, evaporation ramp, 100 days (360 mm)	0.0004 / 0.27 / 0.26 / 0.0002	0.0004 / 0.0005 / 0.0005 / 0.0001
T012, wall, per-cell rate, 50 days	0.0002 / 0.76 / 0.52 / 0.0001	0.0002 / 0.0004 / 0.0004 / 0.0005
Green–Ampt, 24 hourly outputs, against implicit Euler	0.0002 / 0.0009 / 0.0009 / 0.0004	0.0003 / 0.0004 / 0.0004 / 0.0004
Green–Ampt, soil depth 0.2 m (stops at 46 mm)	0.000 / 0.001 / 0.002 / 0.000	exact
Green–Ampt, parameter grids; $K_s = 0$ on one half	identical to numbers; no loss	identical to numbers; no loss
Horton, 24 outputs in 2 h (44.54 mm infiltrated)	0.040 / 0.017 / 0.017 / 0.002	0.040 / 0.017 / 0.017 / 0.053
Volume removed against depth lost	equal (FV2, DG2: within 0.2 %)	equal

ler: up to 0.47 mm with the 60 s steps that all solvers take except CUDA FV1, and 0.94 mm with the steps of 136–607 s that the Courant condition allows CUDA FV1; against the discrete solution for the steps taken, the error is below 0.001 mm. In the Horton test every solver infiltrates the exact 44.54 mm after 2 h to within 0.01 mm (Fig. 1c, d); the larger intermediate differences of Metal ACC reflect output written at the end of the step that crosses each 5 min mark rather than an error of the loss model.

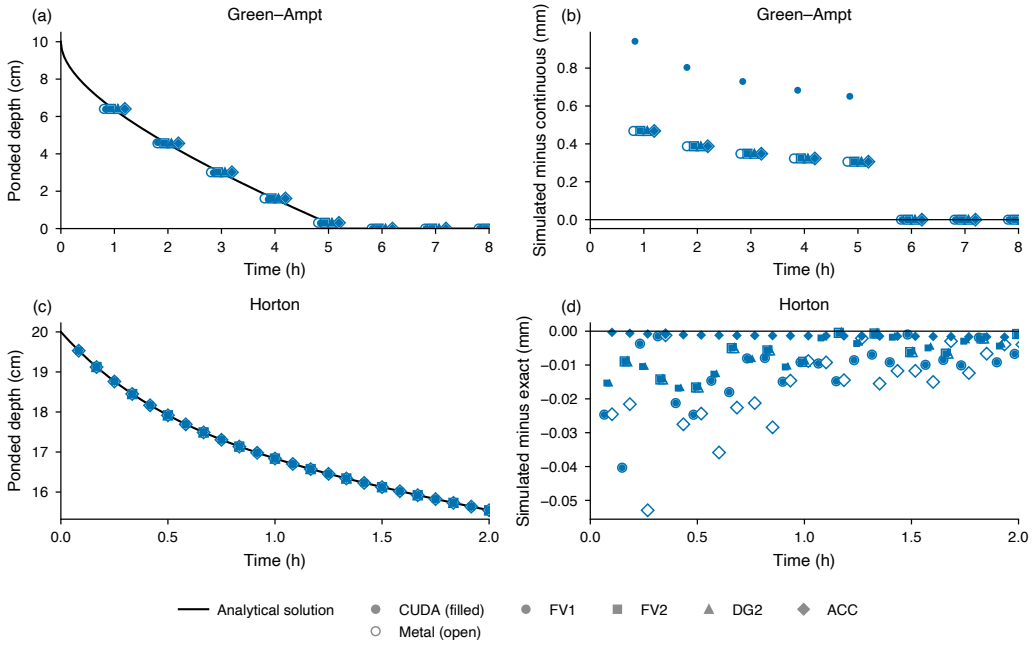


Figure 1. Loss models against analytical solutions, single precision. (a) Ponded depth in the Green–Ampt test and (b) its difference from the continuous solution at the time each grid was written, which is the error of implicit Euler (60 s steps; CUDA FV1 136–607 s). (c) Ponded depth in the Horton test and (d) its difference from the exact solution. Symbols mark the solver, filled symbols CUDA (A100) and open symbols Metal (M1 Max); in (a) all eight runs coincide in value. In (a), (b) and (d) the solvers are drawn slightly apart in time so that every symbol stays visible.

3.2 Free boundary

The boundary correction of Sect. 2.2 is tested on a plane that falls by 2 % away from a free western edge towards a closed eastern edge, with 5 cm of water at the start and no rain. All flow is directed away from the free edge, so no water may enter. Before the correction, Metal FV1 and FV2 gained 0.38 % and 0.39 % of the initial volume in 30 min; after it, FV1 and FV2 conserve the volume to within 0.0003 % on CUDA and Metal, and DG2 to within 0.04 %. The local-inertial and adaptive solvers keep their own rules for the free boundary and are not affected by the correction.

3.3 Consistency with version 8.3

Without the new keywords, 8.4 must reproduce 8.3. On the cases fv1, T01–T03, T030, T031 and T05, every CUDA solver was run twice with each version before the boundary correction: every depth grid was identical between the versions, and the `.mass` logs differed at most in their last digit, as between two runs of the same version. The boundary correction changes results only where the flow at a free edge points into the domain; on the Alzette case of Sect. 4.2, the corrected build reproduces the hourly discharge of the uncorrected build at every gauge, with and without infiltration.

3.4 Losses on adaptive and non-uniform meshes

For the adaptive CUDA solvers (MWDG2, HWFV1) and the non-uniform CUDA ACC solver, analytical basins test constant and per-cell infiltration, evaporation, drying, Green–Ampt with numbers and grids, the soil-depth limit, raster orientation, padding and holes without data, and repeated runs. All 218 checks of the adaptive suite, 5 point-source checks and 106 checks of the non-uniform suite pass. Stored water plus removed volume stays within 1.2×10^{-7} (adaptive solvers, fixed mesh), 2×10^{-5} (adaptive solvers, holes without data) and 2.3×10^{-16} (non-uniform ACC) of the initial volume. On adapting meshes the flow solver itself is not exactly conservative: the volume drifts by 0.1–1.1 % without losses and by 0.2–1.4 % with them. These checks complete the verification of the losses on the non-uniform and adaptive CUDA solvers. The rest of the paper concerns rain-on-grid simulation

Table 4. Metal against the reference over all outputs after the first: share of wet cells (>1 mm) that are identical, share within 1 cm, and largest difference. Reference: CUDA 8.3 for FV2 and DG2, CPU 8.3 for ACC.

Case	FV2	DG2	ACC
T01, T031	identical	identical	identical
T02	identical	99.99 %, 100 %, 1 mm	identical
T03	100 %, 100 %, 1 mm	54 %, 99.3 %, 104 mm	31 %, 45 %, 1.3 m
T030	88 %, 100 %, 1 mm	identical	identical
T05 (0.32 M cells)	99.99 %, 100 %, 4 mm	85 %, 97.6 %, 168 mm	68 %, 80 %, 649 mm
T06 (1.7 M cells, 30 h)	91 %, 99.9 %, 263 mm	76 %, 99.8 %, 120 mm	75 %, 100 %, 35 mm
T07s (0.58 M cells, 6 h)	99 %, 100 %, 2 mm	72 %, 99.6 %, 620 mm[a]	99.5 %, 100 %, 1 mm

[a] First 4 h; afterwards both CUDA and Metal DG2 become unstable on this case (Sect. 6).

of pluvial floods on uniform grids and on several GPUs, for which these solvers are out of scope: from here on, the solvers compared are the uniform FV1, FV2, DG2 and ACC on CUDA and Metal.

3.5 Metal solvers against CUDA and the CPU

Table 4 compares Metal FV2 and DG2 with CUDA 8.3, and Metal ACC with the CPU ACC solver, on the same inputs; the cases contain no inward flow at a free edge, so CUDA 8.3 and 8.4 give the same reference. Metal FV2 is identical to CUDA on T01, T02 and T031, within 1 mm on T03 and T030, and within 1 cm on 99.9–100 % of wet cells of the large cases. Where Metal and CUDA DG2 differ (T03, T05), their first steps agree to four or five significant digits and the difference grows in these sensitive cases; CPU and CUDA DG2 differ by as much (44 % and 84 % of wet cells identical). ACC is similarly sensitive on T03 and T05, and CUDA ACC is unstable on T05 (Sect. 6), so the CPU solver is its reference. On T06, Metal ACC is closer to the CPU solver than CUDA ACC is (75 % of wet cells identical, against 42 % for CUDA). Every Metal run repeats bit for bit. Of the four uniform solvers, FV1 is the one whose Metal port was already compared with CUDA in 8.3, that passes every loss test of Sect. 3.1 on both back ends, and that runs on several GPUs on both (Sect. 3.6); it is the solver used for the Alzette simulations of Sect. 4.2.

3.6 Reproducibility across GPU counts

A run on several GPUs must give the result of a run on one. For CUDA FV1, all saved grids of the four scaling cases of Sect. 5 (10 m and 5 m, up to 14.4 million cells with data, single precision) are identical on one and four GPUs, and so are all 389 grids of each 96 h Alzette simulation of Sect. 4.2 without infiltration, with Green–Ampt and with Horton (double precision). CUDA FV2 and DG2 on four GPUs reproduce the `.mass` record of one GPU on 24 h (FV2) and 6 h (DG2) single-precision runs on the 10 m Alzette grid made for this check. On Metal, two ranks of FV1 on the 10 m Alzette case over 12 h agree with one rank to within 1 mm on 99.45–100 % of wet cells at every hourly output; the largest difference, 16 mm in one cell after 12 h, arises at the rank boundary, where the halo exchange changes the order of floating-point operations.

4 Validation on large real pluvial floods

4.1 Hydraulics: the FloodCastBench reference simulations

FloodCastBench (Xu et al., 2025a) provides inputs and simulated water depths for four pluvial floods: the United Kingdom in December 2015 (3 days), eastern Australia in February and March 2022 (10 days), Mozambique in March 2019 (6 days) and Pakistan in August 2022 (14 days). The reference depths were computed with a local-inertial finite-difference scheme (de Almeida & Bates, 2013) with weighting factor $\theta = 0.7$ and Courant number 0.7, on FABDEM terrain (Hawker et al., 2022) with Manning coefficients from land cover and GPM IMERG rainfall (Huffman et al., 2023). The events are therefore a test of the hydraulic solver on large real domains, with inputs and numerical settings shared with a published model. Two LISFLOOD-FP solvers are run, both on CUDA: ACC, the local-inertial solver of the same family as the reference scheme, and FV1, the full shallow-water solver carried into Sect. 4.2.

The code that generated the dataset is published with it (Xu et al., 2025b), and we used it to fix every detail of the configuration. Re-run on the published inputs, it reproduces the reference frames of Australia and Mozambique with a Nash–Sutcliffe efficiency (NSE; Nash & Sutcliffe, 1970) of 0.99–1.00 and those of the UK with 0.97, averaged over all frames. The model domain is the 30 m terrain

without a margin of 5 cells (UK) or of 150, 50, 50 and 50 cells at the northern, southern, western and eastern edges (Mozambique); Australia uses the whole tile. The 480 m Mozambican reference is the 30 m result sampled bilinearly at evenly spaced points spanning the domain, a spacing of 16.15 and 16.20 cells. Pakistan runs at 480 m on terrain, Manning coefficients and rainfall sampled in the same way, on a domain without margins of 37, 32, 170 and 58 cells, which we located by running the published code over candidate domains and matching the first reference hour. Each rainfall frame is held for its 30 min, Manning coefficients follow the land-cover table of the dataset, missing terrain is raised 30 m above the highest cell, and the fluxes across the domain edges are zero. The simulations start from the first reference frame, except in Mozambique, where the published code starts from a dry domain. In all four references, the stored water equals the initial water plus the rain to within 5 %, consistent with no inflow or outflow and no loss to infiltration or evaporation. In particular, the Pakistani reference contains no inflow from the Indus. With this configuration, the published code reproduces the Pakistani reference less closely (NSE 0.81, already 0.93 after the first hour); we could not identify the cause of the remaining difference. All LISFLOOD-FP runs use the single-precision build on one A100.

The published code also sets every negative flux to zero after each time step, so water in the references can move only eastwards and southwards. Without this restriction, the same code departs from the references with a mean NSE of 0.86 (UK), 0.97 (Australia), 0.67 (Mozambique) and 0.67 (Pakistan). The LISFLOOD-FP solvers let water flow in every direction, so we compare them both with the references and with the published scheme run without the restriction (Table 5, Fig. 2).

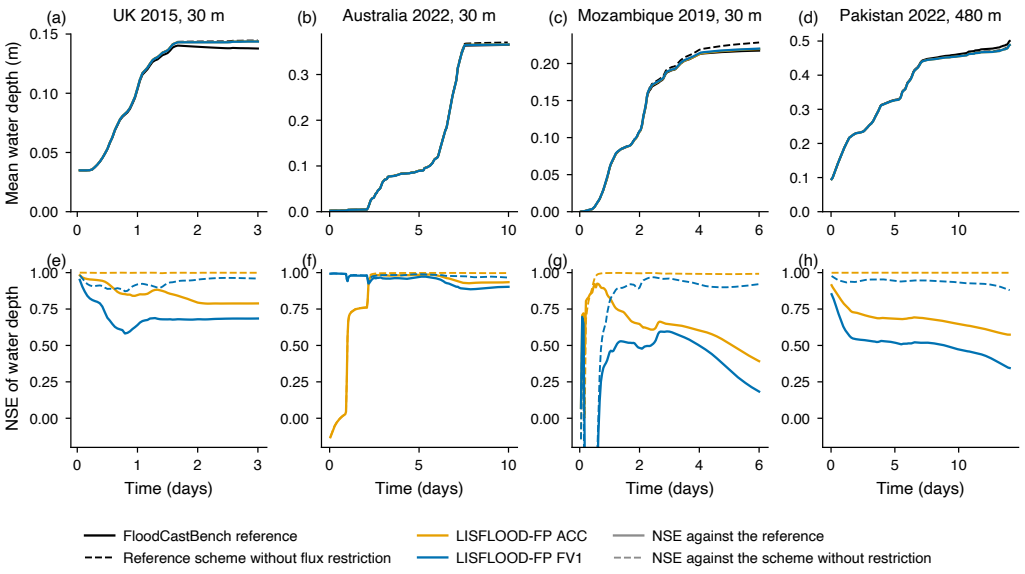
The stored water of every run ends within 4.4 % of the reference (Fig. 2a–d; Table 5). Against the scheme that generated the references, run without the flux restriction, the local-inertial solver of LISFLOOD-FP (ACC) agrees closely: the NSE of the depths is 1.00 in the UK and Pakistan, 0.96 in Mozambique and 0.87 in Australia, and the CSI of the flood extent is 0.94–0.98 (Fig. 2e–h, dashed). The Australian and Mozambican domains are almost dry for the first two days, when small differences in shallow water dominate the measure; from the third day on, the NSE of ACC averages 0.99 in both, against a mean over the whole event of 0.87 (Australia) and 0.96 (Mozambique). The first-order finite-volume solver (FV1), which

Table 5. LISFLOOD-FP 8.4 against the FloodCastBench reference simulations and against the reference scheme without its flux restriction, means over the hourly frames. CSI: critical success index of the cells deeper than 1 and 5 cm. Volume: stored water at the end relative to the reference. Mozambique is compared on the 480 m grid of its reference. Time: wall-clock time of the run on one A100, single precision.

		Against the reference				Without restriction		
Event (cells)	Solver	NSE	$CSI_{1\,cm}$	$CSI_{5\,cm}$	Volume	NSE	$CSI_{5\,cm}$	Time (min)
UK, 30 m (0.05 M)	ACC	0.85	0.80	0.77	+4.4 %	1.00	0.98	0.1
	FV1	0.69	0.55	0.66	+4.3 %	0.93	0.80	0.2
Australia, 30 m (1.15 M)	ACC	0.84	0.94	0.93	+0.3 %	0.87	0.97	1.5
	FV1	0.94	0.90	0.91	+0.2 %	0.98	0.94	3.5
Mozambique, 30 m (5.38 M)	ACC	0.63	0.84	0.80	+1.0 %	0.96	0.94	3.4
	FV1	0.29	0.69	0.71	+1.2 %	0.67	0.80	7.6
Pakistan, 480 m (0.36 M)	ACC	0.68	0.77	0.73	−2.3 %	1.00	0.98	4.1
	FV1	0.51	0.58	0.59	−2.3 %	0.94	0.77	4.2

solves the full shallow-water equations rather than the local-inertial approximation, agrees less closely, with NSE 0.67–0.98 and CSI 0.77–0.94; as the reference scheme and ACC solve the same approximation, these differences measure the gap between the two sets of equations on these domains rather than an error of either solver.

Against the references themselves, the agreement is limited by the flux restriction rather than by the solvers. In Australia, where the restriction changes the reference little, both solvers reproduce it closely: NSE 0.84 (ACC) and 0.94 (FV1), CSI 0.93 and 0.91. In the UK, ACC reaches an NSE of 0.85 against the reference, which is what the unrestricted scheme itself achieves (0.86). In Mozambique and Pakistan the restriction matters more: ACC agrees with the references with NSE 0.63 and 0.68, again close to the unrestricted scheme (0.67 and 0.67), and the agreement falls with time as the restricted flow accumulates (Fig. 2g, h). The FloodCastBench references are therefore a reproducible benchmark for the volume and the broad extent of flooding, but their depths in Mozambique and Pakistan reflect the flux restriction of the generating code. The dataset contains no infiltration or evaporation, so the loss models are evaluated against observations in the next section, with CUDA FV1



388 **Figure 2.** LISFLOOD-FP 8.4 against the FloodCastBench reference simulations. (a–d) Mean
389 water depth over the domain; (e–h) NSE of the water depth against the reference (solid) and
390 against the reference scheme without its flux restriction (dashed).

416 alone, against three other first-order finite-volume CUDA solvers and their own loss
417 models.

418 ### 4.2 Infiltration: the July 2021 flood of the upper Alzette

419 The simulations of this section use CUDA FV1 on one and four A100 GPUs.
420 The three codes it is compared with solve the full shallow-water equations with first-
421 order finite-volume schemes written in CUDA, each with its own infiltration model,
422 so FV1 gives a like-for-like comparison of the loss models. ACC is not used here: the
423 local-inertial approximation neglects the convective acceleration, which is not small
424 in rain-driven flow over the steep valley sides of this catchment, and the comparison
425 would then measure the approximation rather than the losses, as Sect. 4.1 shows for
426 FV1 against a local-inertial reference.

427 *Event and observations* In July 2021, prolonged heavy rain caused a ma-
428 jor flood in the upper Alzette basin in Luxembourg. The basin covers 467.7 km^2
429 on a 10 m grid (4.7 million cells with data in a raster of 4362×2440), with bridge
430 decks removed from the terrain and a gridded Manning coefficient. Simulations

start on 13 July 2021 at 18:00 UTC from a stored water depth and run for 96 h. The rainfall is the gauge-adjusted radar product RADCLIM of the Royal Meteorological Institute of Belgium (Goudenhoofdt & Delobbe, 2016) at 1 km and 5 min, of which 38.7 million m^3 fall on the basin, most between hours 16 and 28. The national water authority observes hourly discharge at four gauges, from the upstream Livange and Hesperange to Pfaffenthal and Steinsel downstream; its records, kept in UTC+1, are shifted to UTC. Each simulation is scored over hours 1–96 at each gauge by three measures. The peak ratio (PR) is the simulated divided by the observed peak discharge, 1 for a perfect peak. The Nash–Sutcliffe efficiency (NSE; Nash & Sutcliffe, 1970) is one minus the sum of squared errors divided by the variance of the observations: 1 for a perfect hydrograph, 0 for a simulation no better than the observed mean, negative below it; the squared errors at the peaks dominate it. The Kling–Gupta efficiency (KGE; Gupta, Kling, Yilmaz, & Martinez, 2009) is $1 - \sqrt{(r-1)^2 + (\beta-1)^2 + (\gamma-1)^2}$, where r is the correlation, β the ratio of the simulated to the observed mean discharge (the bias) and γ the ratio of their coefficients of variation; it is also 1 for a perfect hydrograph, weights timing, volume and variability equally, and any value above -0.41 improves on the observed mean (Knoben et al., 2019). Simulated discharge is the integral of the unit discharge normal to transects across the full wetted valley at each gauge, because at these flows the 10 m model routes part of the water beside the surveyed channel. Scored on the surveyed channel alone, the CUDA FV1 runs of Table 6 give a mean PR of 0.98 and a mean NSE of 0.75 without infiltration and 0.60 and 0.34 with Green–Ampt, because the water routed beside the channel is then not counted (the Horton run was not scored in this way); the full-valley transects are used throughout. The observed peaks rely on rating curves extrapolated beyond the gauged range, so peak ratios close to one lie within their uncertainty.

Other solvers and rules of the comparison To separate the loss models from the hydraulic solver, and to place the cost of CUDA FV1 among comparable codes, the event is repeated with three independent, open-source shallow-water solvers written in CUDA that run on one or several NVIDIA GPUs and solve the same equations with first-order finite-volume schemes: HiPIMS (Xia et al., 2019), which provides Green–Ampt; SERGHEI-SWE (Caviedes-Voullième et al., 2023), which provides Horton; and TRITON (Morales-Hernández et al., 2021), without infiltra-

457 **Table 6.** Discharge at the four Alzette gauges, July 2021, 96 h of radar rainfall, all solvers on
458 the same inputs in double precision: ratio of simulated to observed peak discharge (PR) and
459 Nash–Sutcliffe efficiency (NSE) at each gauge, upstream to downstream, and their means over the
460 four gauges; mean Kling–Gupta efficiency (KGE) over the four gauges; volume infiltrated over
461 96 h where the solver reports it.

		PR					NSE					KGE	Infiltrated
Infiltration	Solver	Liv.	Hes.	Pfa.	Ste.	Mean	Liv.	Hes.	Pfa.	Ste.	Mean	Mean	(10^6 m^3)
None	LISFLOOD-FP	2.47	1.57	1.25	1.72	1.75	−1.01	0.30	0.77	0.43	0.12	0.55	–
	HiPIMS	2.69	1.49	1.24	1.71	1.78	−1.75	0.47	0.80	0.45	−0.01	0.58	–
	TRITON	2.29	1.28	1.20	1.49	1.57	−0.58	0.57	0.81	0.68	0.37	0.53	–
	SERGHEI-SWE	2.29	1.29	1.21	1.50	1.57	−0.60	0.58	0.81	0.68	0.37	0.55	–
Green–Ampt	LISFLOOD-FP	1.14	0.79	0.78	1.12	0.95	0.72	0.67	0.57	0.68	0.66	0.46	14.1
	HiPIMS	1.26	0.79	0.77	1.11	0.98	0.82	0.72	0.61	0.70	0.71	0.62	–[a]
Horton	LISFLOOD-FP	1.04	0.74	0.75	1.08	0.90	0.73	0.64	0.53	0.65	0.64	0.46	14.4
	SERGHEI-SWE	1.84	1.01	0.95	1.28	1.27	0.24	0.63	0.84	0.77	0.62	0.47	6.7

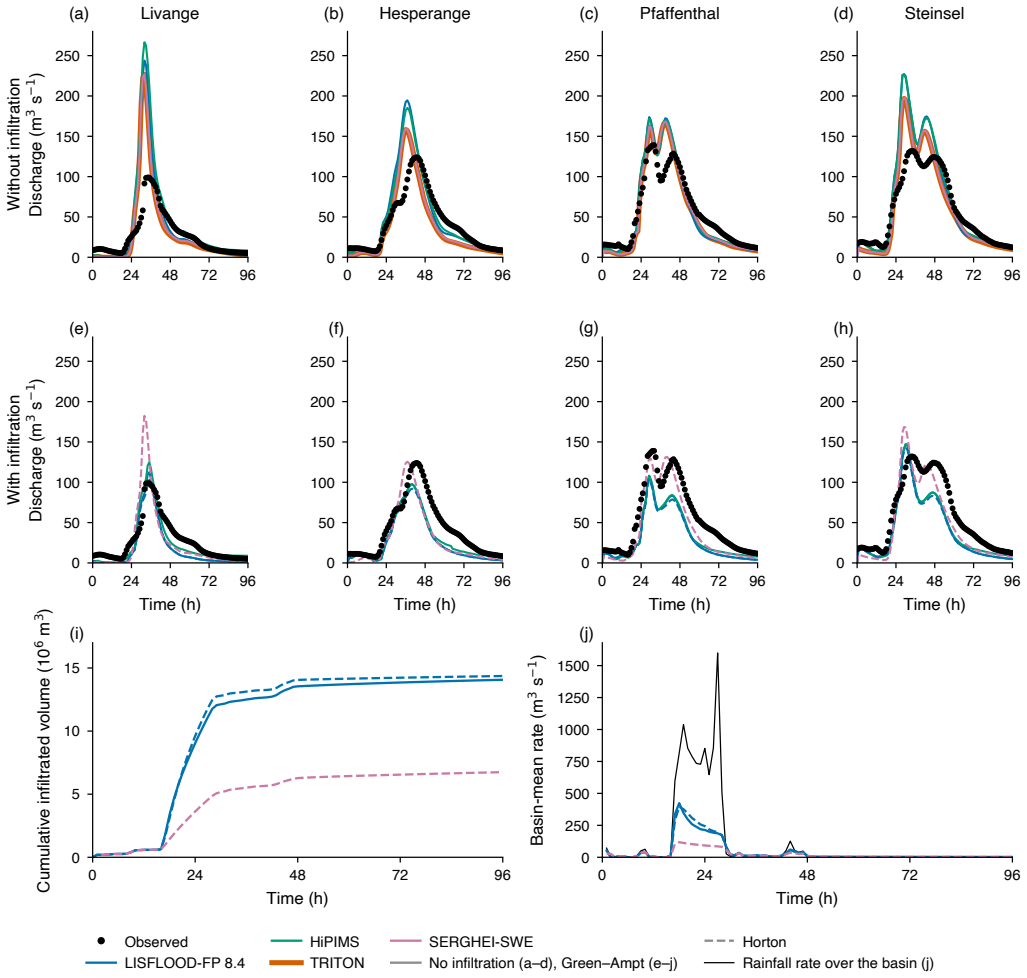
[a] HiPIMS writes no infiltration or water-balance output.

469 tion. They span the two families of Godunov schemes: LISFLOOD-FP FV1 (HLL)
470 and HiPIMS (HLLC, Xia, Liang, Ming, & Hou, 2017) reconstruct the water depth
471 at the cell faces to balance the bed slope, whereas TRITON and SERGHEI-SWE
472 solve augmented Roe problems that contain the bed step. The reconstruction adds
473 operations at every face (Zhao & Liang, 2022), and a Roe solver costs more than
474 HLL or HLLC (Kesserwani et al., 2008); both bear on the run times reported be-
475 low. Every solver receives the same terrain (cells outside the basin without data),
476 Manning coefficients, initial depth, soil parameters where it infiltrates, rainfall (as a
477 rate on the 10 m grid; for TRITON, one runoff series per distinct series of cells, so
478 that every cell receives its rainfall) and outlet, the 25 m section where the Alzette
479 leaves the basin. Each solver represents free outflow in its own way: LISFLOOD-
480 FP removes the water that reaches cells without data, HiPIMS opens the outline
481 of the domain, TRITON treats cells without data as sinks, and SERGHEI-SWE,
482 whose faces to such cells are walls, imposes critical flow on the outlet cells. All four
483 run in double precision and write hourly outputs, on NVIDIA A100-SXM4-40GB
484 GPUs of one node with one MPI rank per GPU, on one and on four GPUs. On four

GPUs LISFLOOD-FP, TRITON and SERGHEI-SWE reproduce their single-GPU discharge exactly; HiPIMS does so to within $0.01\,\mathrm{m^3\,s^{-1}}$ without infiltration and, with Green–Ampt, gives the same peaks to within $0.1\,\mathrm{m^3\,s^{-1}}$ but differs by up to $3.3\,\mathrm{m^3\,s^{-1}}$ at low flow in the first hours. Each solver runs in its documented default configuration, with no option tuned for speed; SERGHEI-SWE uses its implicit centred friction option, because its default friction aborted on this case within the first minute, and exchanges halos through GPU-aware MPI, which its multi-GPU runs require. Run time is the wall-clock time of the solver process, including reading the inputs and writing the outputs, and excluding their conversion between formats. No parameter of any solver was calibrated.

Soil The Green–Ampt parameters come from European soil databases: K_s and the saturated water content of the upper 30 cm from EU-SoilHydroGrids (Tóth et al., 2017), the suction head from the texture relations of Saxton and Rawls (2006), and the initial moisture from SMAP (Entekhabi et al., 2010). Sealed, alluvial and gleyed soils receive $K_s = 0$; the remaining pervious soils cover 71 % of the basin with a mean K_s of $0.76\,\mathrm{mm\,h^{-1}}$. In 781 504 cells (17 % of the basin) the SMAP moisture exceeds the saturated water content of EU-SoilHydroGrids; there $\Delta\theta$ is zero and the soil infiltrates at K_s from the start, in both models. For Horton, the same soil is expressed in Horton's form: in each cell $f_c = K_s$, the rate both models approach once the soil is wet, and f_0 and k_H are fitted by least squares so that the Horton curve of infiltration under ponding matches that of Green–Ampt over the first 24 h (median $f_0 = 5.6\,\mathrm{mm\,h^{-1}}$, median $k_H = 0.125\,\mathrm{h^{-1}}$). Under ponding, Green–Ampt and Horton then infiltrate 25.0 and 24.8 mm after 6 h and 56.7 and 54.3 mm after 24 h (means over the pervious cells), so differences between runs with the two models reflect the form of the model, not the soil.

Hydraulics without infiltration Without losses every solver overestimates every peak, by 20–25 % at Pfaffenthal and by a factor of 2.3–2.7 at Livange (Table 6, Fig. 3a–d), so the overestimate does not come from one solver. The solvers fall into the two families of schemes. LISFLOOD-FP and HiPIMS agree with each other (mean PR 1.75 and 1.78), and so do TRITON and SERGHEI-SWE, whose peak discharges differ by less than 1 % (mean PR 1.57). The second family gives lower peaks at Hesperange and Steinsel, and with them a higher mean NSE (0.37 against 0.12



510 **Figure 3.** Upper Alzette, July 2021; time in hours since 13 July 2021, 18:00 UTC. Hourly
511 discharge at the four gauges, upstream to downstream, observed and simulated: (a–d) without
512 infiltration by the four solvers, (e–h) with Green–Ampt (LISFLOOD-FP, HiPIMS) and Horton
513 (LISFLOOD-FP, SERGHEI-SWE). TRITON (wide line) and SERGHEI-SWE coincide in (a–d).
514 (i) Cumulative volume infiltrated over the basin and (j) its hourly mean rate over the basin, both
515 in $m^3\,s^{-1}$, for the runs that report infiltration: LISFLOOD-FP with Green–Ampt (blue, solid)
516 and with Horton (blue, dashed), SERGHEI-SWE with Horton (purple, dashed); HiPIMS writes
517 none. The thin black line in (j) is the hourly rainfall rate over the basin.

and −0.01), and carries 54–68 % of the low flow of the first family in the first 12 h at Pfaffenthal and Steinsel (5.8 and 7.6 $m^3 s^{-1}$, against 8.6–9.0 and 13.4–14.0 $m^3 s^{-1}$; observed 14.7 and 18.0 $m^3 s^{-1}$). LISFLOOD-FP, TRITON and SERGHEI-SWE store and release nearly the same water: at 96 h, 15.0–15.8 million m^3 remain in the basin and 24.8–25.5 million m^3 have left it, so the schemes differ in how fast shallow water drains along the valley, not in its volume. No solver represents baseflow from groundwater: at Livange all carry 0.3–1.2 $m^3 s^{-1}$ in the first 12 h, against 9.1 $m^3 s^{-1}$ observed.

Green–Ampt run In LISFLOOD-FP, Green–Ampt removes 14.1 million m^3, 36 % of the rain. The outflow falls from 25.5 to 15.5 million m^3, the mean PR from 1.75 to 0.95 and the mean NSE rises from 0.12 to 0.66, with the largest gain at Livange (−1.01 to 0.72; Fig. 3e–h). HiPIMS, with its own implementation of the same model, corroborates the result: its peak ratios differ from those of LISFLOOD-FP by at most 0.01 at Hesperange, Pfaffenthal and Steinsel, and its mean NSE is 0.71. The two differ most at Livange, a wide floodplain gauge where the HiPIMS peak is 11 % higher (9 % without infiltration). The sensitivity to the soil is also consistent: halving K_s raises the mean PR to 1.19 in LISFLOOD-FP and 1.23 in HiPIMS, doubling it lowers the ratio to 0.68 and 0.70, and a wetter initial moisture from ERA5-Land (Muñoz-Sabater et al., 2021) raises it to 1.42 and 1.48. Manning coefficients, the initial state and the rainfall product were not varied, so the uncalibrated scores test the loss models under one parameterisation, not the best attainable fit. Both solvers underestimate the recession and the second peak at Pfaffenthal and Steinsel: infiltrated water leaves the model for good, whereas in the basin part of it returns to the rivers as subsurface flow, and the mean simulated discharge there is 56–72 % of the observed. The two efficiencies therefore move in opposite directions: the NSE, dominated by the peaks, rises, whereas the KGE, which weights the bias in volume as much as the timing, falls in LISFLOOD-FP from a mean of 0.55 to 0.46, because its bias term β drops from 1.0–1.3 at the four gauges without infiltration to 0.56–0.70 with it. HiPIMS keeps a higher KGE (0.62), mainly at Livange, where its discharge is closer to the observed volume.

Horton run and the form of the infiltration model The Horton run of LISFLOOD-FP infiltrates 14.4 million m^3 (37 % of the rain), and its mean PR and

NSE, 0.90 and 0.64, lie within 0.06 and 0.03 of the Green–Ampt run (Table 6). With parameters derived from the same soil, the choice between the two models changes the discharge far less than a factor of two in K_s does. SERGHEI-SWE, given the same f_0, f_c and k_H in every cell (as 34 429 parameter classes rounded to four significant digits), infiltrates only 6.7 million m^3, and its peaks are 19–77 % higher than those of the LISFLOOD-FP Horton run (mean PR 1.27; 1.84 at Livange, with NSE 0.24). The shortfall is confined to the loss term: the water SERGHEI-SWE does not infiltrate leaves the basin (outflow 19.9 against 15.0 million m^3) or remains in it (13.9 against 11.2 million m^3), with a closed balance, as in its run without losses.

SERGHEI-SWE uses Horton's original equation, $f = f_c + (f_0 - f_c)\, e^{-k_H t}$, with t a clock that every cell, wet or dry, advances by the time step from the start of the simulation (`SourceSink.h` of the version used). LISFLOOD-FP uses the cumulative form of Sect. 2.1, in which the capacity falls only as water infiltrates. In the first 12 h the rain is lighter than either capacity, so both solvers infiltrate nearly all of it (0.59 million m^3 each). The heavy rain begins about 16 h after the start; by then the excess $f - f_c$ of the time-based capacity has decayed to about 14 % of $f_0 - f_c$ at the median k_H, also in cells that had stayed dry, whereas the cumulative capacity of those cells is still f_0 and that of the others close to it. During the storm SERGHEI-SWE therefore infiltrates at 0.3–0.5 of the rate of LISFLOOD-FP (Fig. 3j), and by hour 24 it has removed 3.6 against 9.6 million m^3 (Fig. 3i). Applied without flow to the rainfall of 200 000 pervious cells with their own parameters, the two forms alone give the same ratio, 0.38 after 24 h and 0.42 after 96 h, against 0.38 and 0.47 in the simulations, so the difference arises from the form of the model and not from its parameters or set-up. Parameters fitted to a ponded infiltration curve assume that the decay starts when ponding starts; the time-based form starts it at the beginning of the simulation, so its losses depend on when the simulation starts rather than on the water the soil has received. The cumulative form removes this dependence, and under it the Horton and Green–Ampt runs of LISFLOOD-FP agree, as the fit intends. A fairer test of the time-based form, a run started at the onset of the storm or parameters refitted to it, was not made; neither would describe the same soil as the Green–Ampt run. That SERGHEI-SWE reaches a similar mean NSE (0.62) with higher peaks reflects the lower low flows of its scheme family (see the hydraulics without infiltration above) rather than a better match of the losses.

590 **Table 7.** Wall-clock time of the 96 h Alzette simulation in double precision on NVIDIA A100-
591 SXM4-40GB GPUs of one node, one MPI rank per GPU, from the start to the end of the solver
592 process; number of time steps.

Solver	Infiltration	1 GPU (min)	4 GPUs (min)	Speed-up	Steps
LISFLOOD-FP 8.4	none	33.4	23.7	1.41	666 556
	Green–Ampt	33.5	24.1	1.39	661 820
	Horton	34.2	24.7	1.39	662 612
HiPIMS	none	250.2[a]	80.1	3.12	1 726 457
	Green–Ampt	217.2	75.0	2.89	1 767 888
TRITON	none	36.4	13.4	2.72	667 662
SERGHEI-SWE	none	59.7	50.1	1.19	935 374
	Horton	62.7	47.1	1.33	930 337

[a] Elapsed time of the batch job, which also queried the GPU; all other times are those of the solver process, so this time and the speed-up of its row are upper bounds.

593 *Run time* On one GPU, CUDA FV1 of LISFLOOD-FP is the fastest of
594 the four solvers: 33.4 min, against 36.4 min for TRITON, 60–63 min for SERGHEI-
595 SWE and 217–250 min for HiPIMS (Table 7; the 250 min is the elapsed time of the
596 batch job, an upper bound), and also the fastest per step (3.0 ms, against 3.3, 3.8–
597 4.0 and 7.4–8.7 ms). This order is consistent with the cost of the schemes: HLL in
598 LISFLOOD-FP, the Roe solver of TRITON and SERGHEI-SWE, and the HLLC
599 solver with face reconstruction of HiPIMS, which also takes 2.6 times as many
600 steps. The infiltration models add at most 2.4 % to the run time of LISFLOOD-FP.
601 On four GPUs TRITON is fastest (13.4 min) and LISFLOOD-FP second (23.7–
602 24.7 min); Sect. 5 analyses the modest gain of LISFLOOD-FP from four GPUs.

603 5 Computational performance

604 5.1 Optimisations of the CUDA FV1 and ACC solvers

605 Version 8.4 optimises the steps of the CUDA FV1 and ACC solvers, the two
606 solvers used for the real events of Sect. 4. The FV1 step fuses the cell-wise updates,
607 computes the face fluxes in kernels that march along strips of rows, and reduces the
608 time step in the same kernel as the rainfall. The ACC step computes the x and y

discharges in one kernel, fuses the depth update with the copies of the old discharges and the maximum-depth bookkeeping, and reduces the time step on the device, so that the host waits for the GPU only when it writes output. In both solvers the release kernels remain available through an environment variable. The CUDA FV2 and DG2 kernels keep the structure of 8.2 (8.3 added only synchronisation barriers, and 8.4 the losses).

CUDA FV1 and ACC are faster in 8.4 than in 8.3 in all seven real cases tested with no loss keyword, so that both versions compute the same simulation: FV1 on the rain-on-grid 10 m Alzette grid over 24 h from a dry start, and FV1 and ACC on the UK, Australian and Mozambican FloodCastBench inputs. These solvers were run with both versions in single precision on one A100, alternating on the same node, and every output raster of the seven cases is byte-identical between the two versions. Version 8.4 takes 0.30–0.78 of the time of 8.3, that is, it is 1.3–3.3 times faster, for example 2.0 against 5.5 min for FV1 on the Alzette grid (24 h from a dry start, a different run from those of Sect. 5.3) and 0.18 against 0.51 min for the UK event with FV1.

5.2 Apple Metal against CUDA solvers

Table 8 compares the four solvers on the two back ends for the two largest test cases, the torrential valley flood T06 and the fluvial flood T07s, on one A100 and on the GPU of an Apple M1 Max laptop (32 cores, 400 GB s^{-1} of memory bandwidth against 1555 GB s^{-1}). For FV1 the A100 is 3.7–3.9 times faster, in proportion to the bandwidth of the two devices (3.9), as expected for kernels bound by memory traffic (Sect. 5.4). For ACC it is 6.8–7.8 times faster: the CUDA ACC step was fused in 8.4 (Sect. 5.1), whereas the Metal ACC kernels follow the CPU solver (Sect. 2.3). For FV2 and DG2 the laptop is 2.5–5.4 times faster than the A100: the CUDA FV2 and DG2 kernels keep the unoptimised structure of 8.2 (Sect. 5.1), whereas the Metal solvers were written for 8.4 with the structure of the 8.3 Metal FV1 (one shared flux per face, Sect. 2.3). Their step counts agree to within 0.3 %, so the difference lies in the cost of a step, not in the number of steps. Carrying the FV1 kernel structure to CUDA FV2 and DG2 is therefore the largest remaining gain in the CUDA back end.

Table 8. Wall-clock time of the solver process (reading, simulation, writing) for the same case and solver on one NVIDIA A100 (CUDA) and on an Apple M1 Max (Metal), single precision; time per step in milliseconds; ratio of the M1 Max time to the A100 time. Steps: A100 (M1 Max within 0.3 %). T06: Environment Agency test 5, torrential valley flood, 1.7 million cells, 30 h. T07s: Carlisle 2005, fluvial flood, 0.58 million cells, first 6 h. CUDA ACC and FV1 are the solvers optimised in 8.4; CUDA FV2 and DG2 keep the unoptimised kernels of 8.2 (Sect. 5.1).

		A100		M1 Max			
Case	Solver	min	ms step^{-1}	min	ms step^{-1}	Steps	M1 Max / A100
T06	ACC	0.5	0.19	3.3	1.27	154 090	6.8
	FV1	0.8	0.32	3.1	1.16	158 477	3.7
	FV2	45.8	11.9	8.5	2.22	230 264	0.19
	DG2	53.3	12.2	16.6	3.79	262 584	0.31
T07s	ACC	0.1	0.13	0.7	1.03	37 886	7.8
	FV1	0.2	0.20	0.8	0.78	60 611	3.9
	FV2	6.6	4.30	1.9	1.27	91 685	0.29
	DG2	14.9	4.56	6.0	1.81	196 409	0.40

5.3 Scaling of CUDA FV1 on several GPUs for the Alzette basin

The scaling of CUDA FV1 was measured on the 10 m grid of the basin (10.6 million cells, 4.7 million with data) and on a 5 m grid of its upper part (33.9 million cells, 14.4 million with data), over 24 h and 96 h, without infiltration and in single precision; each pair of runs on one and four A100 GPUs ran consecutively on one node. These simulations are not those of Sect. 4.2: they were set up for the multi-GPU tests, before the validation runs, with rainfall from the Weather Research and Forecasting model at 1.3 km instead of the radar product, and they measure the cost of a step, not the discharge. The 10 m simulation over 24 h restarts from a stored state at 18:00 UTC on 13 July and the others start dry at 00:00 UTC on 13 July.

Four GPUs shorten the simulations by a factor of 1.21–1.91 (Table 9, Fig. 4). The efficiency rises with the size of the grid and the length of the simulation, from

Table 9. Wall-clock time of CUDA FV1 on the Alzette basin, single precision, NVIDIA A100 GPUs, without infiltration. 10 m grid: 10.6 million cells, 4.7 million with data; 5 m grid: 33.9 million cells, 14.4 million with data. Restart: from a stored state. Efficiency is the speed-up divided by four. All saved grids are identical on one and four GPUs.

Case	1 GPU (min)	4 GPUs (min)	Speed-up	Efficiency
10 m, 24 h (restart)	4.62	3.82	1.21	30 %
10 m, 96 h	10.77	6.51	1.66	41 %
5 m, 24 h	9.79	5.64	1.74	43 %
5 m, 96 h	55.35	28.97	1.91	48 %

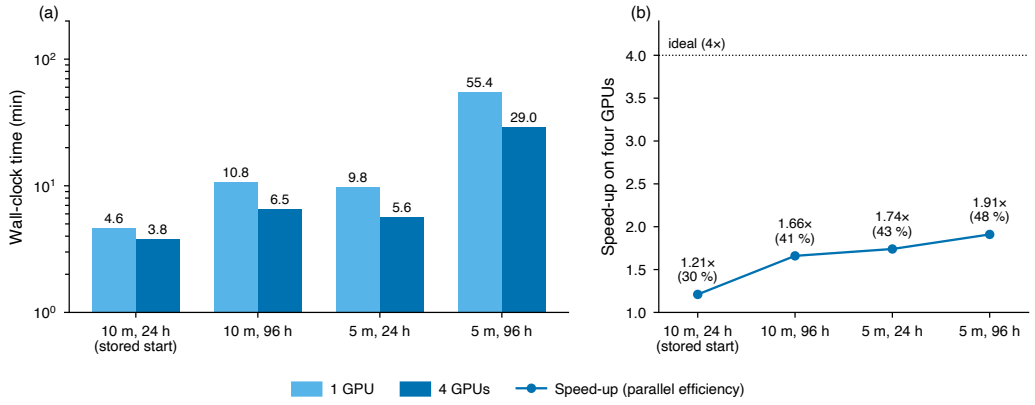


Figure 4. CUDA FV1 on one and four A100 GPUs, Alzette basin, single precision; 10 m grid: 10.6 million cells, 4.7 million with data; 5 m grid: 33.9 million cells, 14.4 million with data. (a) Wall-clock time (logarithmic axis) and (b) speed-up on four GPUs with the parallel efficiency in parentheses; the dotted line marks ideal scaling.

30 % to 48 %, but stays below one half for two reasons. First, a step of 8.4 on one GPU is short: over the 96 h simulations (628 907 and 1 266 801 steps) it takes, in wall-clock time, 1.0 ms on the 10 m grid and 2.6 ms on the 5 m grid, so the fixed cost of each step on several GPUs, the exchange of the halo rows through host memory and the global reduction of the time step, weighs heavily: on four GPUs a step still takes 0.62 and 1.37 ms. Second, the ranks own equal bands of rows, but the wet cells, where the kernels do most of their work, are unevenly distributed among them, as the profiles below show.

687 **Table 10.** Roofline metrics of the three kernels of the CUDA FV1 step on the A100, dou-
688 ble precision, means over 16 launches at a wet state. AI: arithmetic intensity (FLOP byte^{-1});
689 GFLOP s^{-1}: FP64 performance; DRAM and L2: throughput as a percentage of the peak
690 (DRAM 1555 GB s^{-1}). Four GPUs: range of the DRAM throughput over the ranks.

	10 m, 1 GPU				5 m, 1 GPU				4 GPUs, DRAM	
Kernel	AI	GFLOP s^{-1}	DRAM	L2	AI	GFLOP s^{-1}	DRAM	L2	10 m	5 m
Friction	0.42	344	52	68	0.31	262	54	62	29–51	32–54
Flux, x faces	1.75	1238	45	70	1.50	1234	53	76	7–31	27–49
Flux, y faces	0.92	786	55	70	0.80	770	62	73	13–44	40–60

673 ### 5.4 Roofline analysis from hardware counters

674 We profiled the three kernels that carry the work of the CUDA FV1 step (fric-
675 tion and the fluxes and updates across the x and y faces; 2.4 ms of the mean step
676 of 3.0 ms on one GPU at 10 m) with Nsight Compute 2024.3 and placed them on
677 a roofline (Williams et al., 2009). The double-precision build ran the 10 m Alzette
678 simulation of Sect. 4.2 and the 5 m simulation of Table 9, and sixteen consecutive
679 launches of each kernel were replayed and averaged at a wet state: hour 30 of the
680 10 m case, during the flood, and hour 24 of the 5 m case. On four GPUs the launches
681 of the same steps were profiled on every rank. Performance counts the FP64 addi-
682 tions, multiplications and fused multiply–adds (two operations each); the arithmetic
683 intensity divides it by the traffic to device memory (DRAM). The profiles describe
684 double precision, in which a 10 m step takes 3.0 ms (Table 7); the single-precision
685 steps of Table 9 are about three times shorter, with other rainfall and start, so the
686 profiles illustrate rather than measure the balance of that table.

696 The ridge point of the A100 in double precision lies at 6.24 FLOP byte^{-1}
697 (9.7 TFLOP s^{-1} divided by 1555 GB s^{-1}). On one GPU the three kernels reach
698 0.3–1.8 FLOP byte^{-1} and at most 13 % of the FP64 peak, and on four GPUs at most
699 3.6 FLOP byte^{-1} (Table 10, Fig. 5a), so their speed is set by memory traffic rather
700 than arithmetic. On one GPU they use 45–62 % of the DRAM bandwidth and 62–
701 76 % of the L2 bandwidth: the fused kernels take much of their data from the cache,
702 and the metrics hardly change between the two grids. On four GPUs the ranks differ

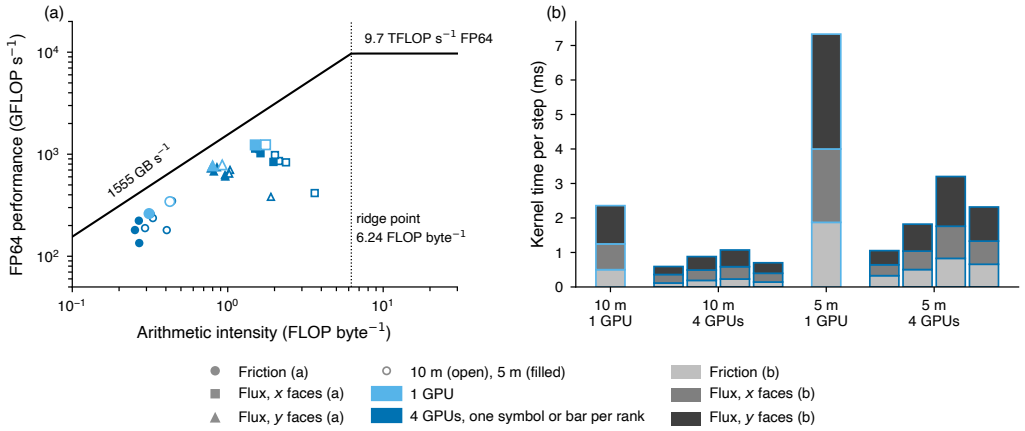


Figure 5. The kernels of the CUDA FV1 step on the A100, double precision; 10 m grid: 10.6 million cells, 4.7 million with data; 5 m grid: 33.9 million cells, 14.4 million with data. (a) Roofline: FP64 performance against arithmetic intensity (filled symbols 5 m, open 10 m); the dotted line marks the ridge point between kernels limited by memory bandwidth and by arithmetic. (b) Time of the three kernels in one step on each GPU: one GPU, and each rank of four.

(Fig. 5b). The kernels of the busiest rank take 1.07 ms per step at 10 m and 3.20 ms at 5 m, against 2.36 and 7.33 ms on one GPU, so that even with free communication four GPUs could run the kernels only 2.2–2.3 times faster. Balancing the ranks by wet cells rather than rows, and exchanging halos directly between GPUs, are therefore the most promising routes to better scaling, and reducing the memory traffic per cell the route to faster steps.

6 Limitations

Neither infiltration model lets the soil drain and regain capacity between rain events (Sect. 2.1), so both suit single events; long simulations require such a recovery, or an initial moisture from a hydrological model. Evaporation is uniform in space and was verified against analytical solutions only; the FloodCastBench and Alzette simulations apply no evaporation. The Horton parameters of the Alzette basin were derived from the Green–Ampt soil, not measured.

The dense Metal modes (Sect. 2.3) provide no non-uniform or adaptive mesh. A six-hour Alzette matrix of 28 runs at 100 m, the seven solver keywords with each loss option, completed on Metal, but the adaptive-grid CUDA solvers were not val-

idated on the basin and are not suited to rain-on-grid simulation: CUDA MWDG2 reads no rainfall input and stalled after 11.37 s with a time step below the resolution of its single-precision clock, and CUDA HWFV1 ignores the dynamic rainfall file and wrote small negative depths.

CUDA FV2 and DG2 on several GPUs reproduce the single-GPU water balance but are fast only when each rank sees only its own GPU (a wrapper script is provided); their device binding has not yet been optimised, and their scaling on full workloads has not been measured. All Metal results come from one Apple M1 Max laptop, whose single 32-core GPU shares 64 GB of memory with the CPU. The row decomposition of Sect. 2.4 runs on Metal with several ranks and was verified with two (Sect. 3.6), but every rank opens the default device, so a computer with several Metal GPUs was not tested. The `.mass` diagnostics of Metal runs with several ranks are not yet correct, although their depth rasters are. The row partition and the halo exchange through host memory limit the scaling of FV1 (Sect. 5.4), and the CUDA FV2 and DG2 kernels, which keep the structure of 8.2, take 4–12 ms per step where the Metal ports take 1–4 (Sect. 5.2).

CUDA FV2 and DG2 keep the bed elevation in single precision as given (Sect. 3.1). The following behaviours of the 8.3 solvers were found while porting and are reported rather than changed: CUDA and Metal DG2 become unstable on T07 after about 4 h and write negative depths on T06; CUDA ACC, whose results 8.4 does not change, is unstable on T05 (CPU and CUDA agree on 6 % of wet cells) and became unstable with evaporation on the 100 m Alzette matrix above, where Metal ACC stayed bounded; on the plane of Sect. 3.2, where no water may cross the free edge, the free boundaries of CUDA ACC and HWFV1, which 8.4 does not change, lose 1.5 % and 21 % of the volume in 30 min (DG2 and MWDG2 0.05 %); the CUDA FV2 slope limiter evaluates minus the inverse of the usual ratio (e.g. Ayog et al., 2021), so it returns a zero slope on monotone profiles and a non-zero slope only at local extrema, where a limiter should return zero; and CUDA applies dynamic rain and point sources with the time step of the previous step. One defect was corrected: CUDA ACC did not wait for its last output before freeing memory (a call commented out in 8.2), which truncated the last output in one run of T03.

7 Conclusions

LISFLOOD-FP 8.4 makes rainfall losses part of every GPU simulation, on CUDA and on Apple Metal, distributes the uniform-grid FV1, FV2 and DG2 over several GPUs and runs the uniform-grid FV1 and ACC 1.3–3.3 times faster than 8.3 (Table 1).

Verification shows that the loss models reproduce analytical solutions to within 1 mm on every solver and both back ends, that a run of 8.4 with no loss keyword reproduces 8.3 exactly, and that CUDA FV1 gives identical results on one and four GPUs. On the FloodCastBench events, set up as the published code that generated the references, LISFLOOD-FP 8.4 ends within 4.4 % of the reference volume, and its local-inertial solver agrees with the generating scheme with an NSE of 0.87–1.00 once that scheme's restriction to eastward and southward flow is removed.

For the July 2021 flood of the upper Alzette, run with CUDA FV1 and three independent first-order finite-volume CUDA solvers on the same inputs, precision and hardware, every solver overestimates the peaks without infiltration, so the losses are needed whatever the hydraulic scheme. Green–Ampt infiltration parameterised from European soil data without calibration removes 36 % of the rain and raises the mean NSE of discharge at four gauges from 0.12 to 0.66, in close agreement with HiPIMS (0.71). A Horton parameterisation of the same soil gives 0.64. The time-based Horton model of SERGHEI-SWE, with the same parameters, infiltrates half as much, because its capacity decays before the storm whether or not the soil has received water; the cumulative form of 8.4 removes this dependence on the start of the simulation. On one A100 GPU, CUDA FV1 completes the simulation of 4.7 million cells with data in 33 min, the fastest of the four solvers. Its kernels are bound by memory bandwidth and its single-GPU step is short, so on four GPUs the uneven wet area of the row partitions and the fixed cost of each step limit the speed-up to 1.2–1.9; balancing the partitions by wet cells and exchanging halos directly between GPUs are the next steps, together with the implementation of the FV1 kernel structure in CUDA FV2 and DG2, which the Metal ports already have (Sect. 5.2).

Acknowledgments

This work was supported by the Fonds National de la Recherche Luxembourg (FNR) under the Industrial Fellowship (IF) grant (Project No. 17130773). During the preparation of this work the authors used Anthropic's Claude models (Opus 5.5 and Fable 5.1) to improve the grammar and language of the manuscript and for AI-assisted coding and debugging of the CUDA and Metal code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. Author contributions: HuR: conceptualization, methodology, software, investigation, formal analysis, visualization and writing (original draft). FNRT: supervision, funding acquisition and writing (review and editing). GK: conceptualization and writing (review and editing). The authors declare there are no conflicts of interest for this manuscript.

Open Research Statement

The source of LISFLOOD-FP 8.4, the patch from 8.3, the tests of Sect. 3, the scripts that set up the FloodCastBench and Alzette simulations and that re-run the FloodCastBench generation code, and the data and scripts behind the tables and figures are [to be archived with a DOI]. LISFLOOD-FP is released under the GNU General Public License v3. FloodCastBench is available from Xu et al. (2025a). The Alzette inputs (Sect. 4.2) are part of the archive; the observed discharge was provided by the Administration de la gestion de l'eau of Luxembourg. The comparison of Sect. 4.2 used HiPIMS (`github.com/HEMLab/HiPIMS-CUDA`, commit 0a27363), SERGHEI-SWE (`gitlab.com/serghei-model/serghei`, commit 2728883) and TRITON (`github.com/lassiterdc/triton`, commit a57ce39); the scripts that convert the inputs for them and score all runs are part of the archive.

References

Ayog, J. L., Kesserwani, G., Shaw, J., Sharifian, M. K., & Bau, D. (2021). Second-order discontinuous Galerkin flood model: Comparison with industry-standard finite volume models. *Journal of Hydrology*, *594*, 125924. doi: 10.1016/j.jhydrol.2020.125924

Bates, P. D., Horritt, M. S., & Fewtrell, T. J. (2010). A simple inertial formulation of the shallow water equations for efficient two-dimensional flood inundation

modelling. *Journal of Hydrology*, *387*(1), 33-45. Retrieved from `https://www.sciencedirect.com/science/article/pii/S0022169410001538` doi: https://doi.org/10.1016/j.jhydrol.2010.03.027

BMT. (2025). TUFLOW classic/HPC user manual, build 2025.1 [Computer software manual]. (Section Green–Ampt infiltration, `https://docs.tuflow.com/classic-hpc/manual/2025.1/`)

Caviedes-Voullième, D., Morales-Hernández, M., Norman, M. R., & Özgen-Xian, I. (2023). SERGHEI (SERGHEI-SWE) v1.0: a performance-portable high-performance parallel-computing shallow-water solver for hydrology and environmental hydraulics. *Geoscientific Model Development*, *16*, 977–1008. doi: 10.5194/gmd-16-977-2023

Chowdhury, A. A., & Kesserwani, G. (2025). LISFLOOD-FP 8.2: GPU-accelerated multiwavelet discontinuous Galerkin solver with dynamic resolution adaptivity for rapid, multiscale flood simulation. *Geoscientific Model Development*, *18*, 9827–9854. doi: 10.5194/gmd-18-9827-2025

de Almeida, G. A. M., & Bates, P. (2013). Applicability of the local inertial approximation of the shallow water equations to flood modeling. *Water Resources Research*, *49*, 4833–4844. doi: 10.1002/wrcr.20366

Dekker, T. J. (1971). A floating-point technique for extending the available precision. *Numerische Mathematik*, *18*(3), 224–242. doi: 10.1007/BF01397083

Entekhabi, D., Njoku, E. G., O'Neill, P. E., Kellogg, K. H., Crow, W. T., Edelstein, W. N., et al. (2010). The Soil Moisture Active Passive (SMAP) mission. *Proceedings of the IEEE*, *98*, 704–716. doi: 10.1109/JPROC.2010.2043918

Goudenhoofdt, E., & Delobbe, L. (2016). Generation and verification of rainfall estimates from 10-yr volumetric weather radar measurements. *Journal of Hydrometeorology*, *17*, 1223–1242. doi: 10.1175/JHM-D-15-0166.1

Green, W. H., & Ampt, G. A. (1911). Studies on soil physics. *The Journal of Agricultural Science*, *4*(1), 1–24. doi: 10.1017/S0021859600001441

Gupta, H. V., Kling, H., Yilmaz, K. K., & Martinez, G. F. (2009). Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. *Journal of Hydrology*, *377*, 80–91. doi: 10.1016/j.jhydrol.2009.08.003

Hawker, L., Uhe, P., Paulo, L., Sosa, J., Savage, J., Sampson, C., & Neal, J. (2022).

A 30 m global map of elevation with forests and buildings removed. *Environmental Research Letters*, *17*, 024016. doi: 10.1088/1748-9326/ac4d4f

Horritt, M. S., Bates, P. D., Fewtrell, T. J., Mason, D. C., & Wilson, M. D. (2010). Modelling the hydraulics of the Carlisle 2005 flood event. *Proceedings of the Institution of Civil Engineers – Water Management*, *163*(6), 273–281. doi: 10.1680/wama.2010.163.6.273

Horton, R. E. (1941). An approach toward a physical interpretation of infiltration-capacity. *Soil Science Society of America Journal*, *5*, 399–417. doi: 10.2136/sssaj1941.036159950005000C0075x

Huffman, G. J., Stocker, E. F., Bolvin, D. T., Nelkin, E. J., & Tan, J. (2023). *GPM IMERG final precipitation L3 half hourly 0.1 degree x 0.1 degree V07*. Goddard Earth Sciences Data and Information Services Center. doi: 10.5067/GPM/IMERG/3B-HH/07

Kahan, W. (1965). Further remarks on reducing truncation errors. *Communications of the ACM*, *8*(1), 40. doi: 10.1145/363707.363723

Kesserwani, G., Ghostine, R., Vazquez, J., Ghenaim, A., & Mosé, R. (2008). Riemann solvers with Runge–Kutta discontinuous Galerkin schemes for the 1D shallow water equations. *Journal of Hydraulic Engineering*, *134*(2), 243–255. doi: 10.1061/(ASCE)0733-9429(2008)134:2(243)

Knoben, W. J. M., Freer, J. E., & Woods, R. A. (2019). Technical note: Inherent benchmark or not? Comparing Nash–Sutcliffe and Kling–Gupta efficiency scores. *Hydrology and Earth System Sciences*, *23*, 4323–4331. doi: 10.5194/hess-23-4323-2019

Liang, Q., & Borthwick, A. G. L. (2009). Adaptive quadtree simulation of shallow flows with wet–dry fronts over complex topography. *Computers & Fluids*, *38*(2), 221–234. doi: 10.1016/j.compfluid.2008.02.008

LISFLOOD-FP developers. (2024). *LISFLOOD-FP v8.2 hydrodynamic model.* Zenodo, `https://doi.org/10.5281/zenodo.13121102`. doi: 10.5281/zenodo.13121102

Mein, R. G., & Larson, C. L. (1973). Modeling infiltration during a steady rain. *Water Resources Research*, *9*(2), 384–394. doi: 10.1029/WR009i002p00384

Morales-Hernández, M., Sharif, M. B., Kalyanapu, A., Ghafoor, S. K., Dullo, T. T., Gangrade, S., ... Evans, K. J. (2021). TRITON: A multi-GPU open source

2D hydrodynamic flood model. *Environmental Modelling & Software*, *141*, 105034. doi: 10.1016/j.envsoft.2021.105034

Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., et al. (2021). ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, *13*, 4349–4383. doi: 10.5194/essd-13-4349-2021

Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models part I – a discussion of principles. *Journal of Hydrology*, *10*, 282–290. doi: 10.1016/0022-1694(70)90255-6

Neal, J., Keef, C., Bates, P., Beven, K., & Leedal, D. (2013). Probabilistic flood risk mapping including spatial dependence. *Hydrological Processes*, *27*(9), 1349–1363. doi: 10.1002/hyp.9572

Néelz, S., & Pender, G. (2013). *Benchmarking the latest generation of 2D hydraulic modelling packages* (Tech. Rep. No. SC120002). Bristol, UK: Environment Agency.

Rossman, L. A., & Huber, W. C. (2016). *Storm water management model reference manual, volume I – hydrology (revised)* (Tech. Rep. No. EPA/600/R-15/162A). Cincinnati, OH: U.S. Environmental Protection Agency.

Saxton, K. E., & Rawls, W. J. (2006). Soil water characteristic estimates by texture and organic matter for hydrologic solutions. *Soil Science Society of America Journal*, *70*, 1569–1578. doi: 10.2136/sssaj2005.0117

Sharifian, M. K., Kesserwani, G., Chowdhury, A. A., Neal, J., & Bates, P. (2023). LISFLOOD-FP 8.1: new GPU-accelerated solvers for faster fluvial/pluvial flood simulations. *Geoscientific Model Development*, *16*, 2391–2413. doi: 10.5194/gmd-16-2391-2023

Shaw, J., Kesserwani, G., Neal, J., Bates, P., & Sharifian, M. K. (2021). LISFLOOD-FP 8.0: the new discontinuous Galerkin shallow-water solver for multi-core CPUs and GPUs. *Geoscientific Model Development*, *14*, 3577–3602. doi: 10.5194/gmd-14-3577-2021

Shirvani, M., Kesserwani, G., & Richmond, P. (2021). Agent-based simulator of dynamic flood–people interactions. *Journal of Flood Risk Management*, *14*(2), e12695. doi: 10.1111/jfr3.12695

Soares-Frazão, S., & Zech, Y. (2008). Dam-break flow through an idealised city.

Journal of Hydraulic Research, *46*(5), 648–658. doi: 10.3826/jhr.2008.3164

Tóth, B., Weynants, M., Pásztor, L., & Hengl, T. (2017). 3D soil hydraulic database of Europe at 250 m resolution. *Hydrological Processes*, *31*, 2662–2666. doi: 10.1002/hyp.11203

Williams, S., Waterman, A., & Patterson, D. (2009). Roofline: an insightful visual performance model for multicore architectures. *Communications of the ACM*, *52*(4), 65–76. doi: 10.1145/1498765.1498785

Xia, X., Liang, Q., & Ming, X. (2019). A full-scale fluvial flood modelling framework based on a high-performance integrated hydrodynamic modelling system (HiPIMS). *Advances in Water Resources*, *132*, 103392. doi: 10.1016/j.advwatres.2019.103392

Xia, X., Liang, Q., Ming, X., & Hou, J. (2017). An efficient and stable hydrodynamic model with novel source term discretization schemes for overland flow and flood simulations. *Water Resources Research*, *53*(5), 3730–3759. doi: 10.1002/2016WR020055

Xu, Q., Shi, Y., Zhao, J., & Zhu, X. X. (2025a). FloodCastBench: A large-scale dataset and foundation models for flood modeling and forecasting. *Scientific Data*, *12*, 431. doi: 10.1038/s41597-025-04725-2

Xu, Q., Shi, Y., Zhao, J., & Zhu, X. X. (2025b). *FloodCastBench: data generation code*. GitHub, `https://github.com/HydroPML/FloodCastBench`, directory Data_Generation_Code, commit c27d108.

Zhao, J., & Liang, Q. (2022). Novel variable reconstruction and friction term discretisation schemes for hydrodynamic modelling of overland flow and surface water flooding. *Advances in Water Resources*, *163*, 104187. doi: 10.1016/j.advwatres.2022.104187